CHAPTER 23
SPACE PHYSICS

6.1 Earth and the Solar System

6.1.1 The Earth

1 Know that:

(a) the Earth is a planet that orbits the Sun once in approximately 365 days

(b) the orbit of the Earth around the Sun is an ellipse which is approximately circular

(c) the Earth rotates on its axis, which is tilted, once in approximately 24 hours

(d) it takes approximately one month for the Moon to orbit the Earth

(e) it takes approximately 500 s for light from the Sun to reach the Earth

2 Define average orbital speed from the equation

$$v = \frac{2\pi r}{T}$$

where r is the average radius of the orbit and T is the orbital period; recall and use this equation

6.1.2 The Solar System

1 Describe the Solar System as containing:

(a) one star, the Sun

(b) the eight named planets and know their order from the Sun

(c) minor planets that orbit the Sun, including dwarf planets such as Pluto and asteroids in the asteroid belt

(d) moons, that orbit the planets

(e) smaller Solar System bodies, including comets and natural satellites

2 Analyse and interpret planetary data about orbital distance, orbital period, density, surface temperature and uniform gravitational field strength at the planet's surface

3 Know that the strength of the gravitational field:

(a) at the surface of a planet depends on the mass of the planet

(b) around a planet decreases as the distance from the planet increases

4 Know that the Sun contains most of the mass of the Solar System and that the strength of the gravitational field at the surface of the Sun is greater than the strength of the gravitational field at the surface of the planets

5 Know that the force that keeps an object in orbit around the Sun is the gravitational attraction of the Sun

6 Know that the strength of the Sun's gravitational field decreases and that the orbital speeds of the planets decrease as the distance from the Sun increases

6.2 Stars and the Universe

6.2.1 The Sun as a star

1 Know that the Sun is a star of medium size, consisting mostly of hydrogen and helium, and that it radiates most of its energy in the infrared, visible and ultraviolet regions of the electromagnetic spectrum

2 Know that stars are powered by nuclear reactions that release energy and that in stable stars the nuclear reactions involve the fusion of hydrogen into helium

6.2.2 Stars

1 State that:

- (a) galaxies are each made up of many billions of stars
- (b) the Sun is a star in the galaxy known as the Milky Way
- (c) other stars that make up the Milky Way are much further away from the Earth than the Sun is from the Earth
- (d) astronomical distances can be measured in light-years, where one light-year is the distance travelled in a vacuum by light in one year

2 Describe the life cycle of a star:

- (a) a star is formed from interstellar clouds of gas and dust that contain hydrogen
- (b) a protostar is an interstellar cloud collapsing and increasing in temperature as a result of its internal gravitational attraction
- (c) a protostar becomes a stable star when the inward force of gravitational attraction is balanced by an outward force due to the high temperature in the centre of the star
- (d) all stars eventually run out of hydrogen as fuel for the nuclear reaction
- (e) most stars expand to form red giants and more massive stars expand to form red supergiants when most of the hydrogen in the centre of the star has been converted to helium
- (f) a red giant from a less massive star forms a planetary nebula with a white dwarf at its centre
- (g) a red supergiant explodes as a supernova, forming a nebula containing hydrogen and new heavier elements, leaving behind a neutron star or a black hole at its centre
- (h) the nebula from a supernova may form new stars with orbiting planets

6.2.3 The Universe

1 Know that the Milky Way is one of many billions of galaxies making up the Universe and that the diameter of the Milky Way is approximately 100 000 light-years

2 Describe redshift as an increase in the observed wavelength of electromagnetic radiation emitted from receding stars and galaxies

3 Know that the light from distant galaxies shows redshift and that the further away the galaxy, the greater the observed redshift and the faster the galaxy's speed away from the Earth

4 Describe, qualitatively, how redshift provides evidence for the Big Bang theory

1. O/N/23/11/Q39

What is the first stage in the life cycle of stars?

A a black hole

B a cloud of gas and dust

C a red supergiant

D a supernova

2. O/N/23/11/Q40

What is the definition of redshift?

A the increase in observed frequency of light from galaxies that are moving away from the Earth

B the increase in observed wavelength of light from galaxies that are moving away from the Earth

C the light emitted by distant galaxies that is blue when it reaches the Earth

D the light reaching the Earth that is red when it is emitted by distant galaxies

3. O/N/23/12/Q40

When observed on the Earth, the redshift of the light from galaxy X is smaller than the redshift of the light from galaxy Y.

Which galaxy is closer to the Earth and which galaxy is receding from the Earth faster?

	closer to the Earth	faster recession
A	X	X
B	X	Y
C	Y	X
D	Y	Y

4. M/J/23/12/Q39

In which region of the electromagnetic spectrum does the Sun radiate the most energy?

A infrared region

B microwave region

C radio wave region

D X-ray region

5. M/J/23/12/Q40

Four of the stages in the life cycle of a star, until it becomes a red giant, are shown.

W Inward force of gravitational attraction is balanced by an outward force from its centre.

X Internal gravitational collapse produces an increase in temperature.

Y It expands.

Z Most of the hydrogen has been converted to helium.

In which order do these stages occur, starting with the earliest?

A $W \rightarrow X \rightarrow Y \rightarrow Z$

B $W \rightarrow X \rightarrow Z \rightarrow Y$

C $X \rightarrow W \rightarrow Y \rightarrow Z$

D $X \rightarrow W \rightarrow Z \rightarrow Y$

6.M/J/23/11/Q39

It takes 27 days for the Moon to orbit the Earth.

The average value of the radius of the Moon's orbit is 3.8×10^8 m.

What is the average orbital speed of the Moon?

A 160 m/s

B 1000 m/s

C 1.4×10^8 m/s

D 8.8×10^8 m/s

7. M/J/23/11/Q40

What is the nuclear reaction that takes place in stable stars?

A fission of helium to produce hydrogen

B fission of hydrogen to produce helium

C fusion of helium into hydrogen

D fusion of hydrogen into helium

ANSWERS

1. **B**
2. **B**
3. **B**
4. **A**
5. **D**
6. **B**
7. **D**

1. O/N/23/21/Q9

The planet Venus orbits the Sun at a constant speed of 3.5×10^4 m/s and takes a time T_V to complete one orbit.

(a) Venus is always 1.1×10^{11} m from the Sun.

Calculate T_V.

T_V = ... s [2]

(b) Explain why speed is a scalar quantity.

....

.... [1]

(c) Velocity is a vector quantity.

(i) Explain what happens to the velocity of Venus as it orbits the Sun.

....

....

....

.... [2]

(ii) There is a resultant force on Venus as it orbits the Sun. The force is perpendicular to the direction of the motion of Venus.

State why this force is needed.

....

.... [1]

(iii) Fig. 9.1 shows the orbit of Venus around the Sun.

On Fig. 9.1, draw an arrow to show the resultant force on Venus.

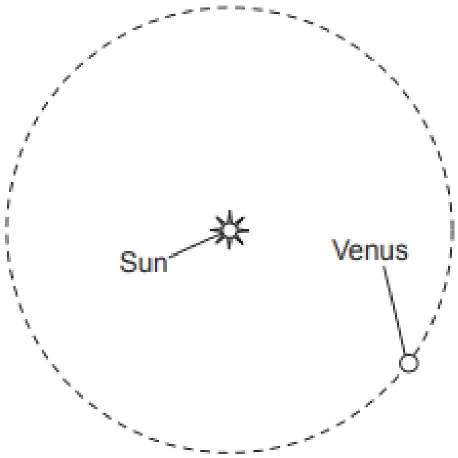

Fig. 9.1

[1]

(iv) State what causes the force acting on Venus.

....

....

.... [2]

(d) One planet in the Solar System is closer to the Sun than Venus.

(i) State the name of this planet.

.... [1]

(ii) Compare the time that this planet takes to complete one orbit of the Sun with T_V and explain the difference.

....

....

....

.... [2]

[Total: 12]

2. M/J/23/22/Q10

(a) Astronomical distances are measured in light-years.

(i) State what is meant by ‘a light-year’.

....

.... [1]

(ii) The Sun is one star in the Milky Way galaxy.

State the approximate diameter of the Milky Way galaxy.

diameter of Milky Way = ... light-years [1]

(b) There are several stages in the life cycle of a star.

(i) Complete Fig. 10.1 to show the stages that a **massive** star goes through after it has used up most of the hydrogen at the centre of the star.

Use words from the following list:

nebula **neutron star** **protostar** **red giant** **supernova** **white dwarf**

massive star (much larger than our Sun)

↓

red supergiant

↓

....

↓

.... and black hole or ....

Fig. 10.1

[2]

(ii) State the stage in the life cycle of a star where heavy elements are formed.

.... [1]

(c) Current scientific understanding is that the universe began 14 billion years ago in an event known as the Big Bang.

Explain **one** observation that supports the Big Bang Theory.

observation ....

....

explanation ....

....

....

....

....

....

[4]

[Total: 9]

ANSWERS

1.

Question	Answer	Marks
9(a)	(T =) $2\pi r / v$ **or** $2\pi \times 1.1 \times 10^{11} / 3.5 \times 10^{4}$	**C1**
	2.0×10^{7} (s)	**A1**
9(b)	it / speed does not have a direction	**B1**
9(c)(i)	it / velocity changes	**M1**
	(because) its direction changes	**A1**
9(c)(ii)	to change the velocity **or** to keep Venus in a circular orbit **or** to accelerate Venus towards the centre of the orbit / Sun	**B1**
9(c)(iii)	arrow towards Sun **and** on Venus	**B1**
9(c)(iv)	it / Venus is in the gravitational field **or** the gravitational attraction	**M1**
	of the Sun	**A1**
9(d)(i)	Mercury	**B1**
9(d)(ii)	time taken is less **and** orbital speed is greater / circumference of orbit is less / closer to the Sun	**B1**
	circumference of orbit is less **and** orbital speed is greater.	**B1**

2.

Question	Answer	Marks
10(a)(i)	distance travelled (in a vacuum) by light in one year	**B1**
10(a)(ii)	100 000	**B1**
10(b)(i)	supernova	**B1**
	nebula **and** neutron star	**B1**
10(b)(ii)	in a supernova	**B1**
10(c)	**either** red shift / increase in (observed) wavelength / reduction in frequency mentioned **or** galaxies / stars moving away (from the Earth) / separating	**B1**
	further away (the galaxy / star) the greater its speed / greater red shift / greater increase in wavelength	**B1**
	going backwards in time / at start) stars or galaxies were close together **or** high density / dense	**B1**
	cosmic microwave background mentioned **or** universe expands	B1
	remnant heat / radiation / left over radiation / radiation from early universe observed now	B1
	radiation has red shifted / longer wavelengths / smaller frequency / become cooler **or** CMBR is uniform / observed in all directions	B1